

"invention" was distinct from the requirement (defined in Art. 57 EPC 1973) of the claimed invention being "susceptible of industrial application". Even though the former requirement might, *cum grano salis*, be equated with a requirement for a "technical" contribution, this was not the same as a requirement for "industrial" applicability; at least in this context, the terms "technical" and "industrial" were not synonyms. In Art. 57 EPC 1973, the meaning of "industrial" was evidently intended to cover commercial applications; this was made clear, for instance, by the German version ("gewerblich"). In the context of Art. 52(2) EPC 1973, this was clearly not the case for the meaning of "technical". This was because, in the context of "controlling a [...] process", the adjective "industrial" should, other than in the context of Art. 57 EPC 1973, be understood to relate only to technical processes usually occurring in "industry" (this latter term in its present context being understood in the narrower sense of what in German would be called "Industrie") and thus not as including "commercial", or even "financial", applications (see also T 208/84, OJ 1987, 14 and T 854/90, OJ 1993, 669).

In T 533/09 the board found that the claimed pulse train was of a tangible nature in that it was the result of modulating an electrical signal (derived from a capacitor discharge for defibrillation purposes) and its intensity was measurable at any time. Such a signal thus clearly came under the definition of "physical entity" within the meaning of G 2/88 (OJ 1990, 93), in its original version. The board concluded that the nature of the subject-matter of claim 1 of the main request was no obstacle to its patentability; moreover, as there could be no doubt that such a signal could be produced and used for defibrillation purposes, that subject-matter clearly was susceptible of industrial application and thus met the conditions of Art. 57 EPC 1973. The board concluded that the wording chosen by the legislator for Art. 52 to 57 EPC on patentability does not in any way link the notion of patentable invention with any condition requiring such an invention to be of a tangible, viz. material, nature (see point 7.2 of the Reasons).

In T 204/93 the board held that commercial use excluded from patentability under Art. 52(2)(c) and (3) EPC 1973 (in so far as it was not construed as a method for doing business) was a feature which might be understood to fall within the requirement that an invention had to be susceptible of industrial application (Art. 52(1) EPC 1973), i.e. that it could be made or used in any kind of industry (Art. 57 EPC 1973); reference was made in particular to the German word "gewerblich" in that provision of the Convention. The board noted that it was not disputed that computer programs could be exploited commercially. The exclusions from patentability defined in Art. 52(2) and (3) EPC 1973 did not in any way refer to the requirement in Art. 52(1) EPC 1973 of "industrial application" but to the requirement in that Article that the subject-matter claimed had to be an "invention". The exclusions in Art. 52(2) and (3) EPC 1973 were generally understood to share the common feature that the excluded matters lacked technicality but not that they could not be made or used, e.g. traded. In this case the board denied patentability for lack of technical effect.

1.2. Any kind of industry

Industry is subject to broad interpretation in this context (T 74/93) and extends to any activity that is carried out continuously, independently and for financial gain (T 144/83) or

that can be offered commercially by a company (T.1165/97). The case law refers to the concepts of "financial (commercial) gain" (T.144/83) and of "profitable use" (T.870/04) in relation to industrial applicability (T.898/05). The need to show a "profitable use" is not to be understood in the narrow sense of an actual or potential economic profit (i.e. generating more income than expenditure) or of a commercial interest (i.e. creating a new or increased business opportunity). Rather, it must be understood in the wider sense that the invention claimed must have such a sound and concrete technical basis that the skilled person can recognise that its contribution to the art could lead to practical exploitation in industry (T.898/05). This interpretation therefore does not extend for example to inventions that merely amount to a research finding (T.870/04, T.338/00) or that are applied for exclusively private and personal purposes.

1.2.1 Methods applied in the private and personal sphere

In T.74/93 (OJ 1995, 712) the application was refused by the examining division because claim 5, which was directed to the use of a contraceptive composition (e.g. a cream) for applying to the cervix of a female capable of conception, was not susceptible of industrial application as required by Art. 57 EPC 1973 in so far as the compound was to be applied to the cervix of a human female.

The board stated that in determining the borderline between industrial activities – in which the effects of patents had to be respected – and private and personal activities, which should not be adversely affected by the exercise of these rights, the board took into consideration the fact that Art. 57 EPC 1973 might be regarded as an expression of the general idea that any natural person had the right to have his or her privacy respected. The core of this right was not to be taken away from anybody. Therefore the fact that for some women contraception was connected with professional activities did not give an act, which was in essence private and personal, an industrial character. The board noted that this did not apply to contraception in general, but to the specific type of application of a composition as claimed in claim 5.

The board was unable to ascertain any field of industrial application for the direct use defined in claim 5, for which the requirement of Art. 57 EPC 1973 had to be met. The question as to whether it would be sufficient for an industrial application to be expected in future could be left unanswered as this assertion had not been further substantiated.

In T.1165/97 the board held that a method of using a vaginal discharge collector and disposing of the collector after a single use could be considered susceptible of industrial application if it was conceivable that these steps were carried out as a paid service and were not exclusively dependent for their execution on the instructions of the woman in question. The board noted that, for the purposes of Art. 57 EPC 1973, what was relevant was the possibility that such a service might be offered by an enterprise. In that case, the board decided in the affirmative, pointing out that – in contrast to the case in T.74/93 – the service was not one satisfying only the strictly personal needs of the woman in question. Collecting a sample could be caused by external reasons, e.g. on the advice of a medical practitioner to have such a sample taken for further diagnosis.

1.2.2 Cosmetic and beauty salons

In T 144/83 (OJ 1986, 301) the board stated that under Art. 57 EPC 1973 an invention was to be considered as susceptible of industrial application if it could be made or used in any kind of industry. Enterprises in the cosmetic field - such as cosmetic salons and beauty parlours - were part of industry within the meaning of Art. 57 EPC 1973, since the notion of "industry" implied that an activity was carried out continuously, independently and for financial gain. One board had already decided in T 36/83 (OJ 1986, 295) that the professional use of such inventions in a cosmetic salon was an industrial application within the meaning of Art. 57 EPC 1973.

2. Reproducibility and sufficiency of disclosure

Another requirement for an invention to be considered susceptible of industrial application within the meaning of Art. 57 EPC is that the technical teaching disclosed in the application under Art. 83 EPC is reproducible.

In T 718/96 the board held that reproducibility was to be assessed in the first instance with reference to Art. 83 and 100(b) EPC 1973, which lay down that the invention must be disclosed in such a way that it can be carried out by a skilled person. Art. 57 EPC 1973 could also serve as a basis for objecting to a lack of reproducibility, because an invention that could not be carried out was also not susceptible of industrial application, but such an objection was always to be based on the most specific applicable provision. Nevertheless, given the link between the two notions, any findings as to reproducibility would apply in equal measure to industrial application under Art. 57 EPC.

In T 1450/07 the board noted that the information in the application as filed should make plausible the identity of the claimed compound. Thus, the compound could be attributed to a known family of molecules on the basis of a comparison between its primary structure and that of molecules known in the art. Then, its putative functions could be disclosed. Experimental evidence was not necessarily needed. A number of reasonable assumptions could be made by taking into account the known functions of other family members as well as, for example and not exclusively, by taking into account the distribution of the claimed compound in the body. The treatments therein mentioned were in relation to the function plausibly attributed to the molecule. Post published evidence backing up these assumptions was always welcome and the quality of the information was also fundamental. The board pointed out that each case had to be evaluated on its own merits. The board decided in the case at issue that the requirement of industrial applicability was fulfilled.

In T 18/09 the board noted that the close inter-relationship between Art. 83 and 57 EPC was already addressed in previous decisions (see inter alia T 898/05). In respect of Art. 83 EPC, established case law of the boards of appeal stated that a patent could only be objected to for lack of sufficiency of disclosure if there were serious doubts, substantiated by verifiable facts (see T 19/90, OJ 1990, 476). For the board, it would not be justified and unfair to set a different standard of proof in respect of Art. 57 EPC.

In T.541/96 the gist of the invention consisted in inducing nuclear fusion between light nuclei and heavy unstable nuclei at low temperature by means of an electric field. The board held that an invention or an application for a patent on an alleged invention which was incompatible with the generally accepted laws of physics did not meet the requirements of Art. 57 and 83 EPC 1973, because it could not be used and therefore lacked industrial application.

The EPC did not exclude "revolutionary" inventions from being patented. However, Art. 83 EPC 1973 made the amount of information required for a sufficient disclosure of an invention somewhat dependent on the actual "nature" of the invention. However, if the invention seemed, at least at first, to offend against the generally accepted laws of physics and established theories, the disclosure should be detailed enough to prove to a skilled person conversant with mainstream science and technology that the invention was indeed feasible (i.e. susceptible of industrial application). This implied, inter alia, the provision of all the data which the skilled person would need to carry out the claimed invention, since such a person, not being able to derive such data from any generally accepted theory, could not be expected to implement the teaching of the invention just by trial and error.

The appellant had provided neither experimental evidence nor any firm theoretical basis which would enable the skilled person to assess the viability of the invention; the description was essentially based on general statements and speculations which were not apt to provide a clear and exhaustive technical teaching. Thus, it was irrelevant to consider whether the fusion reactions referred to in the description might be theoretically possible, or whether they might indeed occur under certain conditions.

3. Indication of a profitable use of the invention in industry

R. 42(1)(f) EPC (former R. 27(1)(f) EPC 1973) prescribes that the description should "indicate explicitly, when it is not obvious from the description or nature of the invention, the way in which the invention is industrially applicable".

In T. 1452/06 the board stated that a basic principle of the patent system was that exclusive rights could only be granted in exchange for a full disclosure of the invention, which included the need to indicate how to exploit the invention (Art. 57 EPC 1973). This indication must have "a sound and concrete technical basis", as a "speculative indication of possible objectives that might or might not be achievable by carrying out further research with the tool as described was not sufficient to fulfil the requirement of industrial applicability" (see T. 898/05, T. 870/04 and T. 1109/10).

In T. 898/05 the board found that for the purposes of Art. 57 EPC 1973, a claimed invention had to have such a sound and concrete technical basis that the skilled person could recognise that its contribution to the art could lead to practical exploitation in industry. The board explained that while industrial applicability should be construed broadly, to include any "immediate concrete benefit", this also entailed the need to disclose in definite technical terms the purpose of the invention and how it could be used in industrial practice to solve a given technical problem, this being the actual benefit or advantage of exploiting the invention. There had to be at least a prospect of a real as opposed to a purely

theoretical possibility of exploitation, if it was not already obvious from the nature of the invention or from the background art. It should not be left to the skilled reader to find out how to exploit the invention by carrying out a research programme.

Accordingly, a product whose structure was given (e.g. a nucleic acid sequence) but whose function was undetermined or obscure or only vaguely indicated might not fulfil the above criteria, in spite of the fact that the structure of the product per se could be reproduced (made) (see [T 870/04](#)). If a patent was granted, it might prevent further research in that area. On the other hand, a product which was definitely described and plausibly shown to be usable, e.g. to cure a rare or orphan disease, might be considered to have a **profitable** use or concrete benefit, irrespective of whether it was actually intended for the pursuit of any trade at all.

In [T 1450/07](#) the board referred to the principles established in [T 898/05](#) that industrial applicability is acceptable if the disclosure in the application as filed on the function of a claimed compound (i) is plausible to the skilled person, (ii) is later confirmed by post-published evidence, and (iii) provides a clear basis for an industrial application.

In [T 870/04](#) the board stated that a "practical" application of the invention had to be disclosed. Merely because a substance could be produced in some ways did not necessarily mean that this requirement was fulfilled, unless there was also some profitable use for which the substance could be employed.

The board noted that biotechnological inventions were quite often concerned with substances found in nature (e.g. a protein, a DNA sequence, etc.). If a function was well known to be essential for human health, then the identification of the substance having this function would immediately suggest a practical application in the case of a disease or condition caused by a deficiency, as was the case, for example, for insulin, human growth hormone or erythropoietin. In such cases, an adequate description would ensure that "the invention can be made or used in industry". In cases where a substance, naturally occurring in the human body, was identified, and possibly also structurally characterised and made available through some method, but either its function was not known or it was complex and incompletely understood; no disease or condition had yet been identified as being attributable to an excess or deficiency of the substance; and no other practical use was suggested for the substance, then industrial applicability could not be acknowledged. Even though research results may be a scientific achievement of considerable merit, they are not necessarily an invention which can be applied industrially.

In cases [T 338/00](#) and [T 604/04](#) the boards found the inventions industrially applicable despite the absence of actual experimental data, because a profitable use could readily be identified on the basis of the description, taking into account common general knowledge. Thus each case is decided on its own merits according to the particular technical circumstances, including extent of disclosure, background art and post-published evidence.

In [T 641/05](#) the board considered that no actual information regarding the function of the CEGPCR1a clone at any of the three particular levels of function referred to in decision